

# IoT Device Management and Operations Platform



## DeviceLive

· Device Management

· Remote Monitoring

· Edge Computing

· Remote Maintenance

# 1. Product Overview

**DeviceLive** is an industrial IoT device management and operations platform. Combined with InHand edge intelligence hardware, it helps industrial enterprises rapidly build smart edge networks. The platform provides four core capabilities: **device management, remote monitoring, edge computing application management, and remote terminal maintenance**. Through cloud-edge collaboration, it enables intelligent edge deployment and upgrades, edge data collection and preprocessing, and visualized status monitoring.

The rapid development of IoT has brought massive numbers of devices and business data. To address digitalization challenges, industrial sectors apply digital and information technologies to achieve **IT/OT convergence**, improving production efficiency, optimizing processes, and enhancing product quality.

## Core Value

**Real-Time Monitoring and Remote Control:** DeviceLive enables real-time monitoring of device status and operation, and supports remote control (such as parameter adjustment and device reboot), improving device reliability and production efficiency.

**Fault Prediction and Maintenance Optimization:** Potential faults can be identified through operational data monitoring, enabling proactive maintenance and reducing downtime and production line interruptions.

**Data Analysis and Decision Optimization:** In-depth analysis of device data provides insights into production processes, optimizes planning and workflows, reduces waste, and improves quality.

**Enhanced Customer Experience:** Supports preventive maintenance, intelligent repair, and fast response to improve customer satisfaction and loyalty.

## 2. License Comparison

| Functionality                  | Basic Edition                                                                              | Advanced Edition                                                                           |
|--------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Device Connection & Management | ✓ Supports device onboarding, configuration management, firmware upgrade and remote reboot | ✓ Supports device onboarding, configuration management, firmware upgrade and remote reboot |
| Remote Access                  | ✓ Supports remote device access for remote maintenance and troubleshooting                 | ✓ Supports remote device access for remote maintenance and troubleshooting                 |
| Device Status Monitoring       | ✓ Supports device status, network status and resource monitoring                           | ✓ Supports device status, network status and resource monitoring                           |
| Data Visualization & Analytics | ✓ Supports device data viewing and basic data analysis                                     | ✓ Supports device data viewing and basic data analysis                                     |
| Alarm Management               | ✓ Supports abnormal alarms for device status, network and resources                        | ✓ Supports abnormal alarms for device status, network and resources                        |
| Report Management              | ✓ Supports device operation data reports                                                   | ✓ Supports device operation data reports                                                   |
| User & Permission Management   | ✓ Supports multi-user and role-based permission management                                 | ✓ Supports multi-user and role-based permission management                                 |

| Functionality                     | Basic Edition                                                                       | Advanced Edition                                                                                                                                             |
|-----------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GPS / Location Management         | ✓ Supports GPS, base station positioning and device location management             | ✓ Supports GPS, base station positioning and device location management                                                                                      |
| InHand DSA Deployment Management  | ✓ Supports InHand DSA deployment management<br>(Applicable to EC/IG series devices) | ✓ Supports InHand DSA deployment management<br>(Applicable to EC/IG series devices)                                                                          |
| Cloud Connection & Networking     | —                                                                                   | ✓ Supports cloud connection and networking capabilities for devices requiring remote network management through DeviceLive                                   |
| Cloud Connection History Analysis | —                                                                                   | ✓ Supports cloud connection history viewing and connection status analysis                                                                                   |
| Edge Application Management       | —                                                                                   | ✓ Supports edge application and containerized application deployment, version management and operation & maintenance<br>(Applicable to EC/IG series devices) |

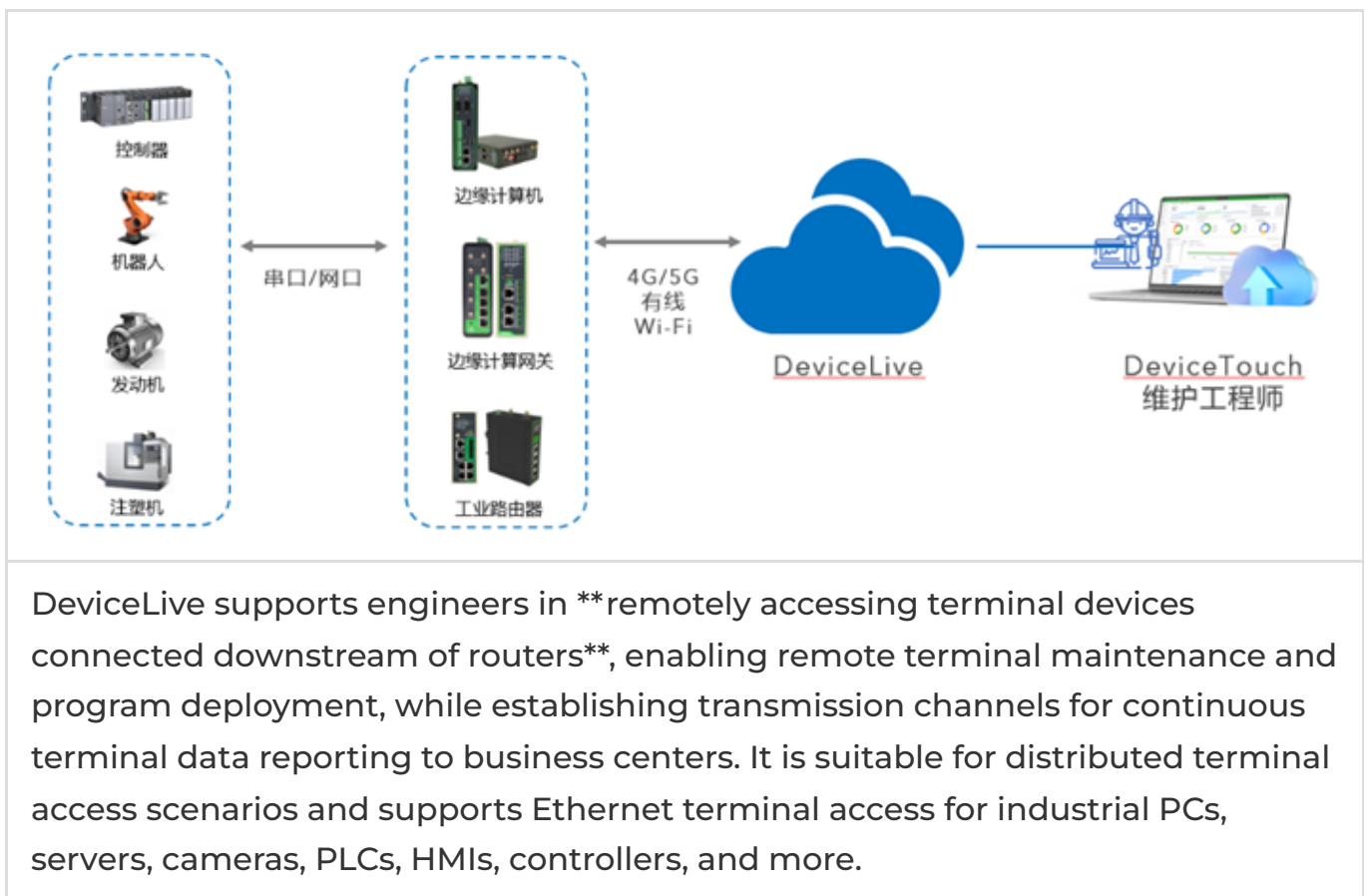
## Plan Description

- **Basic Edition:** Provides basic device onboarding, remote access, device management, status monitoring, data analysis and alarm capabilities for daily device operation and maintenance.
- **Advanced Edition:** Provides advanced cloud connection, edge application deployment and operation & maintenance capabilities based on basic device management, mainly targeting edge computing scenarios.

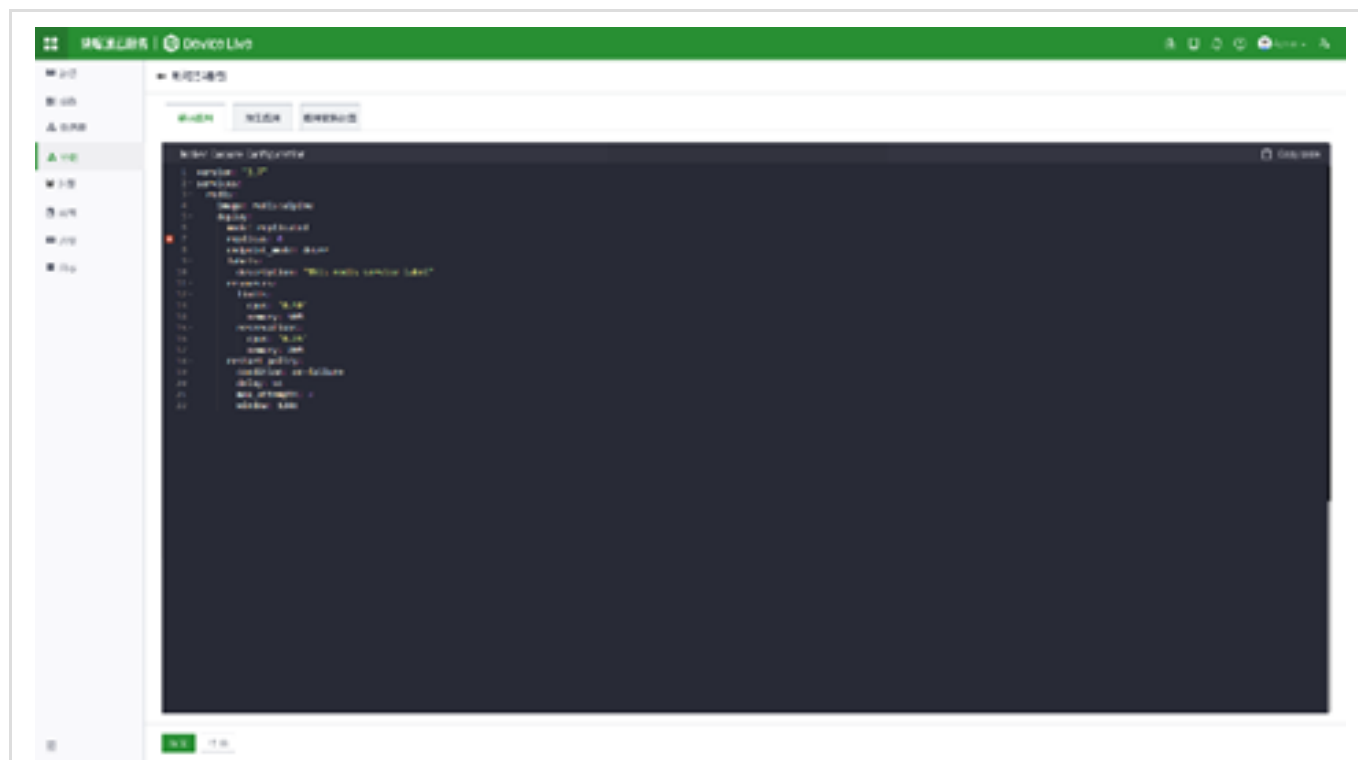
### 3. Platform Capability Overview

DeviceLive provides four core capabilities: **device management**, **network monitoring**, **edge computing application management**, and **remote terminal maintenance**. With cloud-edge collaboration, it enables intelligent edge deployment and upgrades, edge data collection and preprocessing, and visualized status monitoring, comprehensively improving management efficiency.

#### Remote Terminal Access Control



## Edge Computing Application Management



For edge intelligence hardware products, DeviceLive provides management and deployment of edge computing container applications and native applications, without requiring users to build their own OTA services, delivering a one-stop solution.

DeviceLive can centrally configure edge intelligence hardware, manage containers, and upgrade edge computing apps. It supports unified deployment package strategies and rule definitions, enabling centralized upgrades and control of distributed intelligent sites.

## Zero-Touch Deployment and O&M Features

| Capability            | Description                                                   |
|-----------------------|---------------------------------------------------------------|
| Plug-and-Play Devices | Complete configuration and deployment remotely from the cloud |

| Capability                           | Description                                                                               |
|--------------------------------------|-------------------------------------------------------------------------------------------|
| Simple Configuration                 | Web GUI configuration without complex CLI commands                                        |
| Centralized Cloud Management         | Unified device management with batch configuration and upgrades                           |
| Fault Warning and Remote Diagnostics | Multiple alert strategies and remote diagnostics for time-saving troubleshooting          |
| Multi-Dimensional Security Policies  | Data encryption, multi-role permissions, and multi-factor authentication                  |
| Visualized Monitoring                | Comprehensive insights into connection status, network quality, cellular signal, and more |

## 4. Edge Intelligence Hardware and Cloud Data Integration

### Multi-Function Edge Hardware

- Multiple CPU options: from single-core to multi-core ARM processors
- Multiple AI computing power options: approximately 1-26 TOPS, suitable for edge AI scenarios such as facial recognition, speech recognition, and image recognition
- Multiple interfaces: Ethernet, serial, USB, IO, CAN, HDMI, LVDS, etc.

### Distribution Linux and Secondary Development

- Edge intelligence products are built on **distribution Linux**, providing a standard software development environment

- Supports high-level programming languages and Linux community resources to improve development efficiency

## DeviceSupervisor™ Agent

- InHand DeviceSupervisor™ Agent is built in at factory delivery. No programming is required; simple configuration enables **data-to-cloud**
- Supports local data preprocessing to reduce upload traffic and cloud load
- Supports mainstream industrial protocols (such as Modbus, OPC UA, ISO on TCP, etc.)
- Supports integration with major public clouds and IoT cloud platforms, including **AWS, Microsoft Azure, and Alibaba Cloud**

## 5. Application Scenarios

InHand edge intelligence solutions integrate 5G, AI, IoT, and cloud computing, and can be used in **public utility digitalization, predictive equipment maintenance, new energy digitalization, and factory digitalization.**

| Domain                | Description                                                                                                          |
|-----------------------|----------------------------------------------------------------------------------------------------------------------|
| Industrial IoT (IIoT) | Real-time monitoring of device status to improve production efficiency; predictive maintenance to reduce fault risks |
| Smart City            | Smart traffic management and urban digital applications to improve governance efficiency                             |
| Healthcare            | Improved service accessibility with real-time patient status monitoring                                              |
| Energy Management     | Improved energy utilization efficiency, optimized consumption, and reduced waste                                     |

## 6. Platform Feature List

| Feature                           | Description                                                                   | Basic Edition | Advanced Edition |
|-----------------------------------|-------------------------------------------------------------------------------|---------------|------------------|
| Batch Remote Device Configuration | Configure devices remotely                                                    | √             | √                |
| Batch Device Firmware Upgrade     | Remotely upgrade device firmware with flexible scheduled upgrade settings     | √             | √                |
| Device Group Management           | Classify devices by business needs for flexible management                    | √             | √                |
| Remote Control Commands           | Remotely reboot devices and restore factory settings                          | √             | √                |
| Connection Status Statistics      | Monitor device connection status, network type, and more                      | √             | √                |
| Network Status Analysis           | Monitor interface connection status, link status, and traffic consumption     | √             | √                |
| Network Quality Monitoring        | Monitor cellular signal, network latency, jitter, packet loss, and throughput | √             | √                |



|                             |                                                                                                                                                      |   |   |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|
| DSA Management              | Remote DSA configuration, upgrade updates, and status viewing                                                                                        | √ | √ |
| Remote Diagnostic Tools     | Diagnostic logs, Ping, Traceroute, packet capture, and event analysis                                                                                | √ | √ |
| Geolocation Management      | Supports GPS/base-station/manual positioning and map-based device distribution overview                                                              | √ | √ |
| Status Alert Notifications  | Supports multiple alert strategies such as CPU utilization, link status, and cellular traffic monitoring; supports SMS, email, and app notifications | √ | √ |
| Remote Terminal Maintenance | Quickly establish remote channels and support engineer remote access and control of terminal devices                                                 | — | √ |
| Edge Computing Management   | Container management, native application management, and edge computing app upgrade deployment                                                       | — | √ |

|                |                                                                                      |   |   |
|----------------|--------------------------------------------------------------------------------------|---|---|
| MFA            | Multi-factor account authentication for comprehensive security assurance             | √ | √ |
| DeviceLive App | Supports app-based QR code device onboarding and app-based network status monitoring | √ | √ |

Platform URL: [device.inhandcloud.cn](https://device.inhandcloud.cn)

## 7. Contact Us

- Website: [InHand Networks](https://inhand.com)
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